

CLINICAL DENTAL PRACTICE GUIDELINE: LOCAL ANESTHESIA DOSING, CALCULATIONS, AND TOXICITY PROTOCOL

Department of Oral and Maxillofacial Surgery & Dental Therapeutics

Guideline Code: DENT-ANESTH-2024-V2

1. COMMONLY USED LOCAL ANESTHETICS IN DENTISTRY

A. Lidocaine HCl 2% with 1:100,000 Epinephrine

- Concentration: 2% lidocaine = 20 mg/mL. Standard dental cartridge contains 1.8 mL = 36 mg of lidocaine hydrochloride.
- Vasoconstrictor: Epinephrine 1:100,000 = 0.01 mg/mL = 0.018 mg epinephrine per 1.8 mL cartridge.
- Maximum Recommended Dose (MRD) - Healthy Adult: 4.4 mg/kg (or 2.0 mg/lb), with an absolute maximum ceiling of 300 mg (approximately 8.3 cartridges for a 70 kg adult).
- Pediatric MRD: 4.4 mg/kg strictly calculated based on actual body weight.
- * Example: For a 20 kg child: $20 \text{ kg} \times 4.4 \text{ mg/kg} = 88 \text{ mg}$ max lidocaine. $88 \text{ mg} / 36 \text{ mg per cartridge} = 2.4$ cartridges maximum.
- * Example: For a 30 kg child: $30 \text{ kg} \times 4.4 \text{ mg/kg} = 132 \text{ mg}$ max lidocaine. $132 \text{ mg} / 36 \text{ mg per cartridge} = 3.6$ cartridges maximum.
- Onset: 2 to 4 minutes. Pulpal anesthesia duration: 60 minutes. Soft tissue anesthesia: 3 to 5 hours.

B. Articaine HCl 4% with 1:100,000 or 1:200,000 Epinephrine

- Concentration: 4% articaine = 40 mg/mL. A 1.8 mL cartridge contains 72 mg of articaine hydrochloride.
- Maximum Recommended Dose: 7.0 mg/kg (3.2 mg/lb) up to an absolute maximum ceiling of 500 mg.
- High lipid solubility provides superior bone penetration (useful for mandibular buccal infiltration in molars).
- Caution: Higher reported risk of paresthesia with mandibular nerve blocks; recommended primarily for maxillary infiltration and mandibular infiltration.

C. Mepivacaine 3% Plain (Without Vasoconstrictor)

- Concentration: 3% mepivacaine = 30 mg/mL = 54 mg per 1.8 mL cartridge.
- Maximum Recommended Dose: 4.4 mg/kg, absolute max 300 mg (approximately 5.5 cartridges).
- Indication: Patients for whom epinephrine is contraindicated (severe cardiovascular disease, uncontrolled hyperthyroidism, sulfite allergy).

2. VASOCONSTRICTOR LIMITS IN MEDICALLY COMPROMISED PATIENTS

- Healthy Patient (ASA I): Maximum epinephrine dose per appointment is 0.2 mg (up to 11 cartridges of 1:100,000).
- Cardiac / Cardiovascular Compromise (ASA III or IV):
 - * Maximum epinephrine dose per appointment is restricted to 0.04 mg (equivalent to MAXIMUM TWO cartridges of 1:100,000 epinephrine).
 - * Always aspirate in two planes before injection to prevent accidental intravascular administration.
 - * Contraindicated in patients with unstable angina, recent myocardial infarction (< 6 months), refractory arrhythmias, or uncontrolled hypertension (BP > 180/110 mmHg).

3. SIGNS OF LOCAL ANESTHETIC SYSTEMIC TOXICITY (LAST)

- Early CNS Excitation: Metallic taste in mouth, circumoral numbness, tinnitus, lightheadedness, restlessness, visual disturbances, slurred speech.
- Progression: Muscle twitching, tremors, generalized tonic-clonic seizures.
- CNS & Cardiovascular Collapse: Respiratory depression, bradycardia, hypotension, ventricular arrhythmias, cardiac arrest.
- Emergency Management:

1. Terminate dental procedure immediately.

2. Call Emergency Medical Services (911).

3. Maintain airway, administer 100% oxygen via non-rebreather mask.

4. In severe LAST with hemodynamic instability, administer 20% Intralipid emulsion (1.5 mL/kg initial bolus).